

Designing climate-resilient agriculture through crop portfolios

Abstract (1000 characters): Hotter, drier summers and increasing extremes are challenging growers across the world. This is especially clear in Europe where summer heat is increasing faster than in the Arctic. Maintaining agricultural regions that continue to provide economic livelihoods, high-quality products while continuing traditional practices requires understanding of the climate resiliency of current crops, and how to best exploit it. Here I propose to leverage the deep agriculture knowledge, data and cutting-edge research in Montpellier with my skills in hierarchical modeling to address this. Working with unparalleled data across varieties and clones of one major crop, we will quantify heat tolerance across cultivars and clones to help growers develop climate-smart portfolios.

Background

Developing resilient agricultural systems today requires human adaptation to deal with the increasing stressors of climate change. Hotter, drier summers and increasing extremes in heat, cold and precipitation are challenging growers across the world, especially in Europe where summer heat is increasing faster than in the Arctic. Maintaining agricultural regions that continue to provide economic livelihoods, high-quality products while continuing traditional practices relies on a greater understanding of the climate resiliency of current crops, and how to best exploit it as the climate continues to shift. [1]

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More on the importance of heat tolerance and heat tolerance across crops and cultivars.

Then more on why we do not have good information on this to help growers adapt: the current hurdles this project will overcome.

What?

Working with colleagues in Montpellier, I will develop temperature response curves for the crop of winegrapes including its cultivars and clones. We plan to use winegrapes as a case study because they currently have the data on seasonal development and genetic information required to build the most robust models, but our approach will be specifically designed to extend to other crops.

How?

Our approach centers around (1) the domestication of winegrapes across Europe's extensive climate gradient that led to cultivars (varieties) that likely vary strongly in their heat tolerance and (2) the shared evolutionary history of winegrape cultivars that we can use for more robust estimates across diverse varieties. We will accomplish this by leveraging existing climate data with Domaine de Vassal's incredible collection of seasonal development data across over 200 varieties, alongside likely other data from resources across INRAe and other regional programs that have collected data on season development, from budburst to flowering, and fruit ripening (veraison).

Collaborators in Montpellier will provide expertise in the seasonal development data across cultivars and their genetic relationships, which we will model by combining two unique modeling approaches my lab has developed. The first is a fully differentiable thermal development curve (Fig. fig:tempcurve), and the second is a using an hierarchical approach that leverages genetic relationships between cultivars to best estimate the heat tolerance of each cultivar. Each approach has successfully been applied in other cases to woody plants, but have not been combined to estimate thermal responses across diverse cultivars. Given sufficient information on clones (within cultivars) we will additionally add this hierarchical level to separate out the climate adaptation potential of

clones versus cultivars.

Timeline: Data gathering is already underway, with the goal of starting work on model design and parameterization in March 2028 when I arrive in France. In my first month I plan meetings to review and refine my modeling plan with my collaborators, who bring unique expertise in variety diversity, its genetics and how to make results most immediately useful to growers. Model development and fitting will be an iterative process focused on design, parameterization, validating and eventually forecasting. I expect continual consultations with my French collaborators and likely two months of discussion and effort for fitting and validation. In May I hope to begin using the model to prepare historical results, followed by forecasting with the model in June and finalizing results in July.

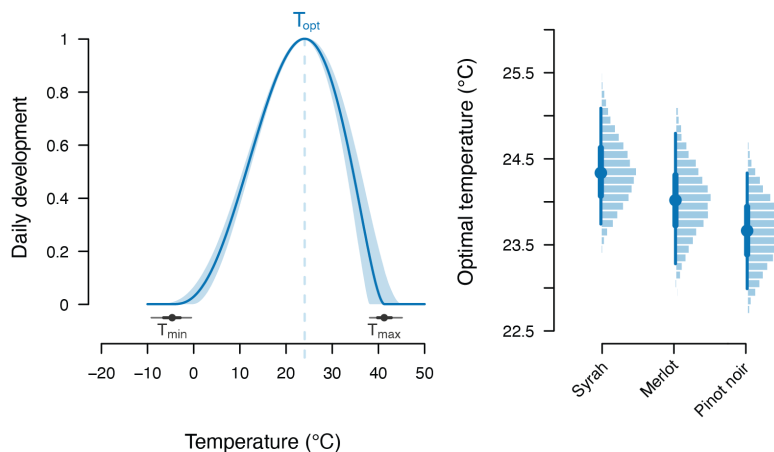


Figure 1: This project will estimate a temperature response curve for winegrapes as a crop, including heat tolerance variation across cultivars. Here we show current results for veraison thermal development (left) and variation in the optimal temperature for three cultivars (right) but this project will extend to up to 200 different cultivars (exact number depends on data for each phenological stage) and more accurately model them given genetic information.

Impact, outcomes & benefits

Our aim is to provide one major building block to climate-smart agriculture: robust estimates of the thermal tolerance of a crop, and its underlying genetic levels, to help growers develop a portfolio approach to future climate. This information should allow growers to build portfolios of varieties that can deal with current climate means and extremes, and help them decide when to shift this portfolio as the climate shifts in the future.

My research on winegrapes to date has generated high interest, both in research and public spheres. For example, my lab’s recent work on variety change with warming [2], has been cited almost 50 times in one year and led to dozens of radio and print stories, including by the BBC, NPR, *The Guardian*, *Washington Post*, *El Mundo* and France24 News. I plan to communicate results through publications (I expect at least one paper in *Nature Climate Change* or similar, one methods and conceptual paper in a high-profile climate change focused journal), press releases and then at various conferences and seminar series I am involved with, including both in academic (e.g., universities), grower-focused (e.g., Napa County Farm Bureau, BC Winegrape Council) and mixed venues (e.g., international Cool Climates Wine Symposium).

Importantly, I expect this research will also open new lines of communication. Climate change is a global challenge and solutions will come from shared problem solving and resilience. My hope is this research will help develop new links with growers and researchers in Europe and in turn foster deeper connections with their counterparts in North America.

Give a talk in Montpellier? Or other outreach opportunities?

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- [1] Coupel-Ledru, A., Westgeest, A. J., Albasha, R., Millan, M., Pallas, B., Doligez, A., Flutre, T., Segura, V., This, P., Torregrosa, L., et al. (2024). Clusters of grapevine genes for a burning world. *The New phytologist*.
 - [2] Morales-Castilla, I., de Cortazar-Atauri, I. G., Cook, B. I., Lacombe, T., Parker, A., van Leeuwen, C., Nicholas, K. A., and Wolkovich, E. M. (2020). Diversity buffers winegrowing regions from climate change losses (vol 117, pg 2864, 2020). *Proceedings of the National Academy of Sciences of the United States of America* *117*, 30860–30860.